Questions

Q1.

The Sun is a star that is in a stable state called the main sequence.

What will the Sun become at the end of its main sequence?

(1)

☐ **A** a black hole

☐ **B** a nebula

☐ **C** a red giant

☐ **D** a supernova

(Total for question = 1 mark)

Q2.

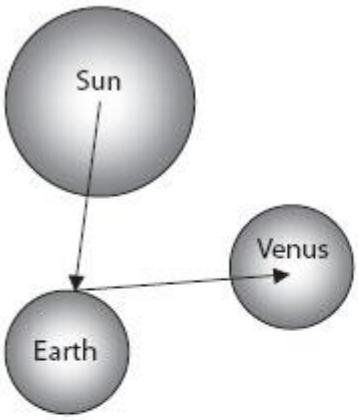
(a) Galileo was one of the first scientists to use a telescope to study Venus.

(i) Which of these diagrams best shows how light waves enable us to see Venus?

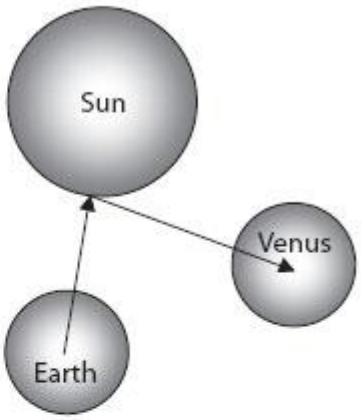
Put a cross ( ) in the box next to your answer.

(1)

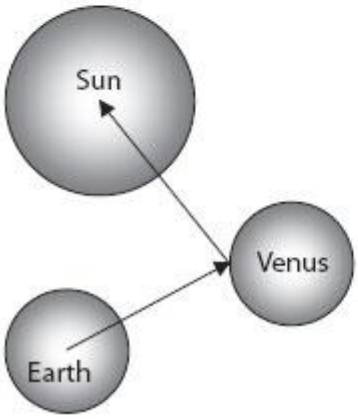
Not to scale



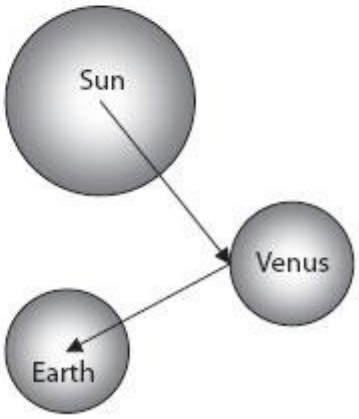
☐ A



☐ B



☐ C



☐ D

(ii) Use words from the box to complete the sentences.

(2)

asteroids	comets	geocentric	heliocentric
	moons	particle	stars

Galileo also used his telescope to observe the ..... of Jupiter.
His observations provided evidence to support the ..... model of the Solar System.

(iii) Describe how a reflecting telescope is different from the simple telescope which Galileo used.

(2)

.....

.....

.....

.....

(b) (i) The diagram represents a wave.

State the amplitude and wavelength of the wave.

(2)

.....

.....

(ii) 20 waves are sent out in 4 seconds.

Complete the sentence by putting a cross ( ) in the box next to your answer.

The frequency of the wave is

(1)

☐ **A** 0.2 Hz

☐ **B** 5 Hz

☐ **C** 20 Hz

☐ **D** 80 Hz

(Total for Question is 8 marks)

Q3.

Explain why Mercury orbits the Sun in a much shorter time than Neptune orbits the Sun.

(2)

.....

.....

.....

.....

.....

(Total for question = 2 marks)

Q4.

Some visible light telescopes are located in space.

Other visible light telescopes are located on the Earth's surface.

Explain why the images produced by telescopes on Earth are less clear than the images produced by telescopes in space.

(2)

.....

.....

.....

.....

Q5.

CMB radiation provides evidence that the Universe had a definite beginning.

Use the table in Figure 20 to give a typical value for the wavelength of CMB radiation.

type of radiation	typical wavelength
gamma rays	1.0×10^{-12} m
X-rays	3.0×10^{-11} m
ultraviolet	200 nm
visible	600 nm
infrared	4.0 μm
microwaves	1.0 mm
radio waves	50 m

Figure 20

(2)

wavelength = ...

(Total for question = 2 marks)

Q6.

Figure 9 shows some lines in the absorption spectra from four different galaxies (A, B, C, and D) and from a laboratory source.

All the spectra are aligned and to the same scale.

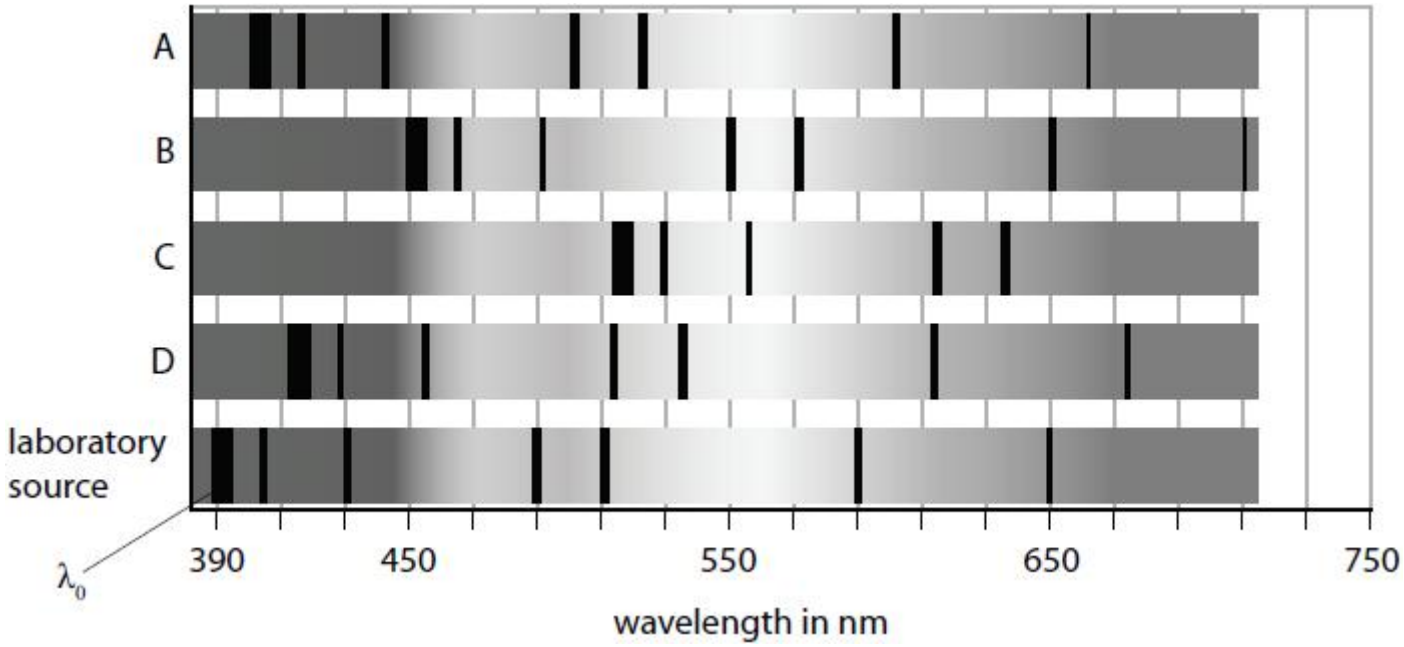


Figure 9

Explain, using Figure 9, which galaxy is furthest away from us.

(3)

.....

.....

.....

.....

.....

.....

(Total for question = 3 marks)

Q7.

During the twentieth century red-shift and CMB radiation were discovered.
They have provided scientists with data to test theories of the origin of the Universe.
(a) (i) Complete the following sentence.

(1)

.....

(ii) State which theory about the origin of the Universe is supported by the existence of CMB.

(1)

.....

(iii) There is a red-shift in the light received from some galaxies.
State what is meant by red-shift.

(1)

.....

.....

(iv) Some galaxies show greater red-shift than others.
Explain what this suggests about the Universe.

(2)

.....

.....

.....

.....

(b) Stars have different stages in their evolution.
(i) Which of these gives the next stages in the evolution of the Sun?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** white dwarf then black hole

☐ **B** neutron star then white dwarf

☐ **C** red giant then supernova

☐ **D** red giant then white dwarf

(ii) Modern telescopes can provide us with more data than the telescopes used 100 years ago.
Explain what additional data can be collected and processed using modern telescopes.

(2)

.....

.....

.....

.....

(Total for Question = 8 marks)

Q8.

The value of g (gravitational field strength) on Earth is 10 N/kg.

On the Moon g is 1.6 N/kg.

An object weighs 10N on Earth.

What would it weigh on the Moon?

(1)

☐ **A** 0.16 N

☐ **B** 1.6 N

☐ **C** 16 N

☐ **D** 60 N

(Total for question = 1 mark)

Q9.

Which row of the table shows these objects in the correct order of size?
Put a cross (☒) in the box next to your answer.

(1)

%%%	**smallest** → **biggest**		
☐ **A**	Milky Way	Solar System	Universe
☐ **B**	Milky Way	Universe	Solar System
☐ **C**	Solar System	Universe	Milky Way
☐ **D**	Solar System	Milky Way	Universe

Q10.

Put a cross (☒) in the box next to your answer.
Which of these is the biggest?

(1)

☐ **A** the Solar System

☐ **B** a galaxy

☐ **C** a nebula

☐ **D** the Universe

Mark Scheme

Q1.

Question number	Answer	Mark
	C red giant	(1)

Q2.

	Answer	Acceptable answers	Mark
(a)(i)	D		**(1)**
(a)(ii)	moons (1) heliocentric (1)	must be in correct order	**(2)**
(a)(iii)	A description including two of the following points Reflecting telescope has mirror(s) (1) Galilean telescope has only lenses (1) Reflecting telescope can gather more light / can have a larger objective (1) Image viewed from the side of reflecting telescope (1) Image viewed from end of Galilean telescope. (1)	refracting telescope reverse argument	**(2)**
(b)(i)	5 (cm) (1) 8 (cm) (1)	+5 -5 0.08 m 80 mm	**(2)**
(b)(ii)	B		**(1)**

Q3.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification - understanding (1 mark) and reasoning/justification - understanding (1 mark): radius of (circular) orbit smaller (1) length of the orbit of Mercury is smaller (1)	 Mercury is closer to the Sun Mercury travels faster	(2)

Q4.

	Answer	Acceptable answers	Mark
	An explanation linking any two of (presence of Earth's) atmosphere (1) causes light to be absorbed/reduced in intensity (1) causes distortion of the image(1) (more) **light** pollution (1) (bigger) variations in temperature (1)	Accept reverse argument (more) air/ clouds/ pollution/ dust blocked / (more) difficult to see through blurs the image / refracts the light	**(2)**

Q5.

	Answer	Additional guidance	Mark
	1(.0) (1) mm (1)	Allow values between 1.0 and 1.9 allow 1×10^{-3} m or 0.001 m for 2 marks if nothing in answer line, credit answer indicated in table	**(2)** **AO3**

Q6.

Question number	Answer	Mark
	An explanation that combines identification via a judgement (1 mark) to reach a conclusion via justification/reasoning (2 marks): • galaxy C is furthest away (1) • because it has the greatest red shift (1) • and therefore it has the greatest speed (1)	**(3)**

Q7.

	Answer	Acceptable answers	Mark
(a)(i)	cosmic microwave background (radiation) all three words needed		**(1)**
(a)(ii)	(the) Big Bang (theory)		**(1)**
(a)(iii)	change in the (observed) frequency **or** wavelength of light/radiation (received from a distant galaxy)	light/absorption lines is/are shifted toward red end of spectrum light/radiation has longer wavelength/lower frequency waves are more spread out/stretched Ignore references to sound or colour of light eg galaxies/light appear(s) red(er)	**(1)**
(a)(iv)	An explanation linking the following:- It/the Universe is expanding (1) some galaxies are (moving)faster (than others) (1)	they/ galaxies are moving further/away (from the Earth/us) the further away the galaxy is the greater the red-shift/the faster the galaxy is moving (some galaxies) are moving away faster (than others) gains 2 marks IGNORE references to planets/stars	**(2)**
(b)(i)	D red giant then white dwarf		**(1)**
(b)(ii)	an explanation linking **two** of the following: • Different/more wave(length)s/frequencies (now) detected/collected (1) • because telescopes positioned above Earth's atmosphere or by radio telescopes (1) OR • weaker signal(s)	named type of em radiation accept CMB for microwave space telescope or named space telescope (because) less or no light/radiation is absorbed by Earth's atmosphere Accept named space telescope eg Hubble/Planck/Compton etc Ignore references to clearer	**(2)**

	(now) detected/collected (1) • because modern telescopes are more powerful/have greater magnification **or** positioned above Earth's atmosphere/ on top of mountains (1) If no other marks awarded allow 1 mark for idea that: Electronic(s)/computers can process/improve the data/signal information/waves/radiation	images/more detail/can see further/photographs	

Q8.

Question number	Answer	Mark
	B 1.6 N	(1)

Q9.

	Answer	Acceptable answers	Mark
	D		**(1)**

Q10.

	Answer	Acceptable answers	Mark
	D the Universe (1)		**(1)**